

Class Notes: Class Chondrichthyes

Animal Kingdom • PU-I Biology • Class Notes

Cartilaginous Fish Features

Marine habitat; cartilaginous endoskeleton, streamlined body; notochord persists throughout life. Skin has small placoid scales; predaceous with strong jaws, mouth located ventrally, teeth are modified placoid scales directed backward. Respiration by 5-7 pairs of gills without operculum; must swim continuously to avoid sinking (no air bladder); heart two-chambered (1 auricle, 1 ventricle).

Physiology and Examples

Some have electric organs (e.g. Torpedo, 8-220 volts) or poisonous stings (e.g. Trygon/sting ray); poikilothermous. Sexes separate; males have claspers on pelvic fins; internal fertilisation; many are viviparous. Examples: Scoliodon (dog fish), Pristis (saw fish), Carcharodon (great white shark), Trygon (sting ray).